



A low-cost, open-source single-photon detection system

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Motivation

- Educational outreach



- Hardware experimentation



- Cost constraints



This was made **at home**.

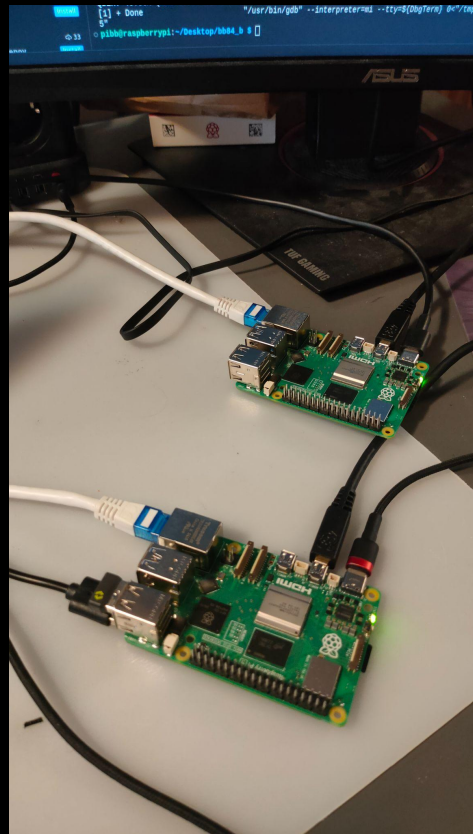
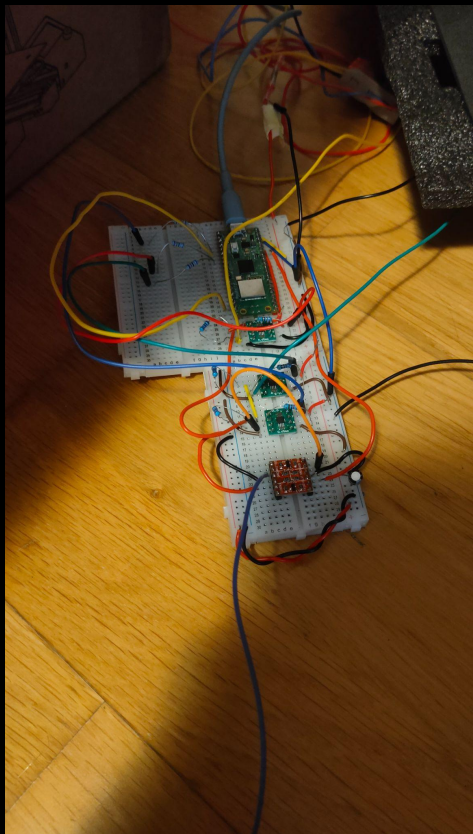


This project was **fully self funded**.

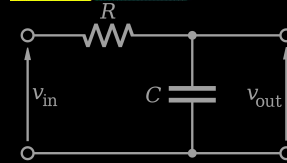
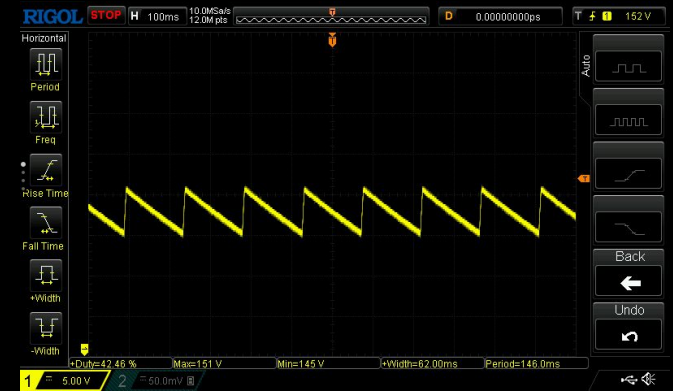
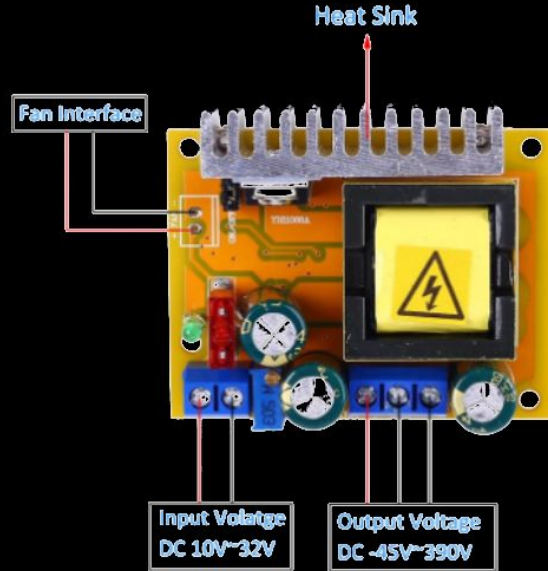


Open-source, so you can try it yourself.



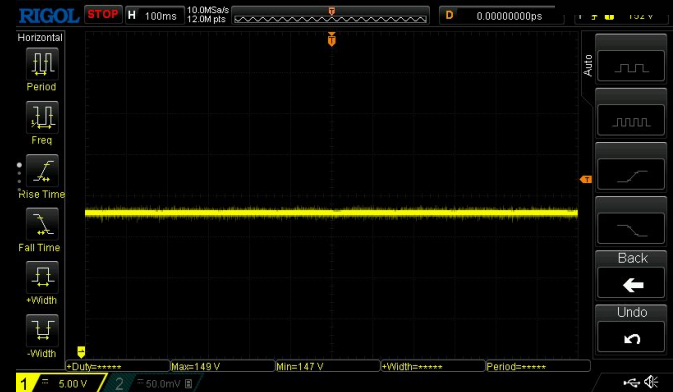


Hardware Design of the Adjustable Boost Converter

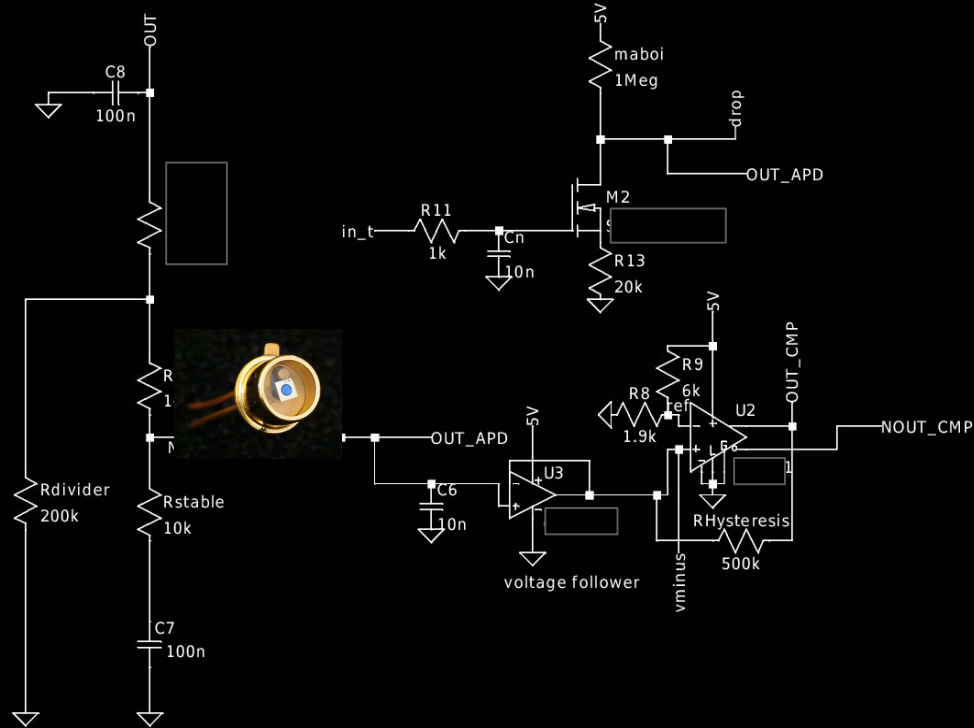


Low pass filter

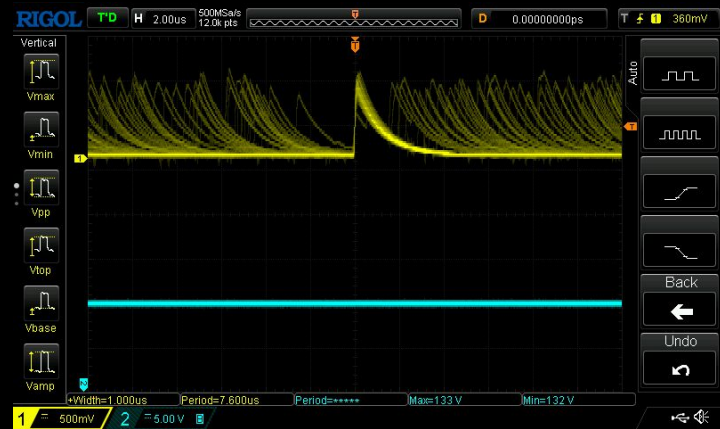
Cut-off ≈ 0.16 Hz



Hardware Design of the Single Photon Detector with Active Quenching



Results



Results

average of 40% qubit error rate

$$\text{QBER} = \frac{\text{Number of mismatched bits}}{\text{Total number of sifted bits}}$$

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SERIAL MONITOR
Connected to Pi A
Final key (sifted):
1011000001110101101011100011011010111010100010000111110001101000001001010101010101100101101011101000110001101100110100111000010011010100110001011101010101
001010011011110010001110101100010111010101010101000100100000110100001001010010
Corrected key:
0011000001000101110100011100011011010111010100010000000100011010000010010101000101010010010110101010100011000110110011010100011000010011010100110001011101010101
0010100010111000100011101011000101010100101010101010100010010000010010000001010010
QBER: 40.56% (101 errors out of 249 sifted bits)
[1] + Done "/usr/bin/gdb" --interpreter=mi --tty=${DbgTerm} 0<"/tmp/Microsoft-MIEngine-In-ulqtmstg.54k" 1>"/tmp/Microsoft-MIEngine-Out-glygbeyb.2b
5"
pibb@raspberrypi:~/Desktop/bb84_b $
```

Ln 60, Col 1 Spaces: 4 UTF-8 LF {} C